

Opinion as Filter, Evidence as Verdict: The Mathematics of Isegoria, an Authority-Free Mechanism for Validating Test Items

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Describes the reference implementation at commit `2ee5e79` of
<https://github.com/DavideMacchia/isegoria>

Abstract

Who decides whether a quiz question is fair when no institution is credibly impartial? Majority voting hands control to the largest faction, and naive reputation systems hand it over faster. Isegoria is a design for producing banks of test items without a trusted authority. It rests on three pillars: externally certified uniqueness of anonymous participants, a peer-review filter that rewards approval *across* a latent divide (“bridging”) instead of approval *by many*, and a final verdict computed from response data with item response theory and differential item functioning (DIF) detected on latent classes, so that no personal attribute is ever collected. This paper states the mechanism formally and analyses it. We establish five results, each with a proof or a reproducible experiment. (i) In the bridging model with an unpenalized global mean, item intercepts sum to zero at every stationary point, so the acceptance gate ranks items against their own batch. In the reference simulation, five of six consensus items pass when fitted alongside four divisive items, and none of the six passes when they are fitted alone. (ii) The origin of the latent axis is a gauge that only the regularization fixes. In a two-camp model the intercept interpolates between the camp-balanced and the camp-size-weighted mean approval with weight $1/(1 + \rho)$, $\rho \approx (\lambda_b/\lambda_f)\sqrt{S/n}$. At the default penalties, 53–87% of the majoritarian effect leaks into the bridge score for n between 50 and 3,200 reviewers. (iii) Latent-class DIF is identified through conditional covariance between biased items: a single biased item carries 250 to 300 times less information than a biased pair, which explains why isolated distortions are invisible. Measurement error in the ability proxy creates spurious latent classes. On null batches the production detector flags clean items when ability is estimated from 10 or 20 anchors, and passes at 30 anchors by a margin of 0.01–0.06. (iv) As is known for skill scores, the ratio-form Brier skill score used for evaluators is not proper. With one scored item and a crowd baseline b , the optimal report satisfies $\text{logit } p^* = \text{logit } q + 2 \text{logit } b$, which pushes dissenters toward the crowd. A leave-one-out difference score is strictly proper and still gives exactly zero to a reviewer who copies the crowd. (v) The sublinear anti-collusion discount is only as strong as its detector, and at design scale two reviewers share under one item per epoch, so random assignment, not the discount, carries the per-epoch defence. We state the evidence level of every claim, list what remains open, and give an evaluation plan.

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1 Introduction

1.1 The problem

Many social processes rest on a bank of test items: civic-knowledge questionnaires, professional certifications, licensing examinations, the claims checked by a fact-checking service. Whoever writes and selects the items shapes what the test measures, and in many settings no institution is credibly impartial. A government that selects the questions of a civic test can select questions that flatter it. A professional body that writes its own licensing exam has a stake in who passes it.

The obvious decentralized alternative is to let participants propose items and vote on them. Voting moves control from the institution to whichever faction holds the majority. Reputation systems in which approval earns weight make matters worse: a cohesive group gains weight faster than a dispersed one and then uses it. The hard part is not collecting items. It is deciding which items are fair *without trusting a single authority and without turning the decision into a popularity contest*.

1.2 The mechanism in brief

Isegoria¹ is a design for this problem [29]. It rests on three pillars, in order of importance.

1. **Uniqueness without identification.** Participants are anonymous, but each anonymous participant is backed by exactly one real person. Uniqueness is certified by an external identity system and then cryptographically severed from everything the person does. Without it no aggregation rule survives a Sybil attack [14].
2. **A quality signal that is not an opinion.** Items are trialled on samples of respondents, and the final verdict comes from psychometrics on the response data: item response theory (IRT) for discrimination and difficulty, and differential item functioning (DIF) for bias [23, 28]. Because no attribute of any person is ever collected, DIF is detected on *latent* classes.
3. **Bridging instead of majority.** Peer review is aggregated by a matrix factorization that credits approval only to the extent that it is not explained by a latent axis of division, in the style of Community Notes [50]. An item approved only by the largest camp is filtered out.

Peer review (Level A) is only an upstream filter deciding which items are worth trialling. The verdict is empirical (Level B). Reviewer reputation (Level C) weights the review and is scored against outcomes, with a proper scoring rule so that honest forecasting is the best strategy. A sublinear discount limits coordinated blocs. Because bridging cannot tell a true but divisive fact from partisan propaganda, an author may appeal a polarization rejection directly to the evidence filter, staking reputation. No money enters the system at any point.

1.3 Contributions

This paper states the mechanism formally and analyses it. The analysis is the main contribution. Several of its results expose properties that the design documents do not anticipate. We report them as findings with proposed corrections, not as settled features of the design.

1. **Formal statement** (Section 2). Requirements, adversaries and a set of assumptions that abstract the identity and network layers. The analysis therefore holds for any realization of

¹Greek *isēgoria*: the equal right of every citizen to speak in the assembly.

those layers that meets the assumptions.

2. **Level A** (Section 3). We characterize the invariances of the bridging objective (Lemma 1) and derive three consequences. Bridge scores sum to zero within a fit, so the gate is relative to the batch (Proposition 2). The origin of the latent axis, which decides whether the score is camp-balanced or majoritarian, is fixed only by the regularization; in a two-camp model we obtain the interpolation weight in closed form (Proposition 4) and confirm it on the full model (Table 3). Finally, no rule that reads only ratings can separate true-but-divisive items from partisan ones (Proposition 6).
3. **Level B** (Section 4). We show why latent-class DIF is identified by the conditional covariance between biased items, and why a single biased item is nearly invisible (Propositions 7 and 8). We also show that measurement error in the ability proxy produces spurious latent classes (Proposition 10), and we measure its effect on the production detector.
4. **Level C** (Section 5). The ratio-form Brier skill score is not proper, a known property of skill scores that here reverses the intended incentive (Proposition 12). A leave-one-out difference score is proper and keeps the property that matters for the design, namely zero reward for following the crowd (Proposition 14). As currently parameterized, the weight cap never binds (Proposition 15).
5. **Coordination** (Section 6). The sublinear discount rewards evading its detector (Proposition 17). At design scale reviewers share almost no items within an epoch (Proposition 18), so the per-epoch defence is random assignment.
6. **Evidence and plan** (Sections 7 to 9). The evidence level of every claim, the structural limits, and an evaluation plan with decision rules fixed in advance.

1.4 Status and reading guide

This is a design-and-analysis paper about a research-stage system. It does not claim that the mechanism works in the field. No component has been externally reviewed, every operational threshold is provisional, and the identity and network layers are partly scaffolds. To keep claims honest, results carry one of four labels, following the verification methodology of the repository [29]: [proved] (a proof is given), [empirical] (a simulation result, valid in the stated regime only), [design] (a choice that is motivated but not derived), and [open] (a question we cannot yet answer). Every number in the paper can be regenerated with the scripts listed in Appendix C. Where we report the behaviour of the production code, we run the Rust engine itself rather than a re-implementation.

2 Setting, requirements and threat model

2.1 Actors and roles

A *person* enrolls once and receives an anonymous credential. From it the person derives exactly one pseudonym for each of three roles:

- **author**: proposes items;
- **evaluator**: reviews items proposed by others;
- **respondent**: answers trial items, which are mixed with validated ones and do not count toward the respondent’s own score.

The three pseudonyms of one person are unlinkable to each other and to the person. A *consortium* of independent, always-on signer nodes stores the append-only record of items and

reviews, recomputes the scores and signs the result. Anyone may run a light node that verifies signatures. A threshold *issuing committee*, distinct from both the consortium and the state identity provider, issues credentials. The state authenticates a person but never issues the credential, so it cannot link a person to their activity.

2.2 Requirements

The following requirements are the invariants of the design [29]. Every mechanism below is evaluated against them.

Requirement R1 (Anonymity). No personal, demographic or affiliation data enters the system, not even optionally.

Requirement R2 (No majority rule). Acceptance of an item is never decided by counting votes.

Requirement R3 (No money). There is no token, stake or payment. Proposing costs reputation and is rate-limited.

Requirement R4 (Separated reputations). The author score and the evaluator score live on unlinkable pseudonyms and are never combined.

Requirement R5 (Non-rotatable pseudonyms). A person can derive exactly one pseudonym per role, so a bad reputation cannot be discarded [18].

Requirement R6 (Reproducibility). The scores are a deterministic function of the recorded inputs, identical bit for bit on every recomputation.

Requirement R7 (Batch validation). Empirical validation is performed on groups of items, never on a single item.

2.3 Adversaries

Table 1 lists the adversaries the design addresses and where each is analysed. The adversary that motivates the whole design is the *majority faction*: a large, cohesive group of real persons that votes sincerely but one-sidedly. It needs no fraud to capture a voting system. A *cartel* is a coordinated bloc of real persons. A *Sybil* adversary creates fake identities. A *long-con* reviewer builds reputation in order to spend it later. A *biased author* writes items whose wording favours one group.

2.4 Assumptions that abstract the identity and network layers

The scoring mechanism is the subject of this paper. The cryptographic identity layer and the replicated storage layer enter only through the following assumptions, so the analysis does not depend on how they are realized.

Assumption A1 (Uniqueness and unlinkability). Each person holds at most one credential. Each credential yields exactly one pseudonym per role. Pseudonyms of different roles are computationally unlinkable to each other and to the person’s identity.

Assumption A2 (Unpredictable assignment). Admission, reviewer assignment, honeypot placement and sortition use public randomness that no participant can predict or bias before the inputs it applies to are fixed.

Assumption A3 (Blind simultaneous review). Every reviewer of an item commits to a judgment before any judgment is revealed. Reviewers see neither the author nor the other judgments.

Table 1: Adversaries, primary countermeasures, and where this paper analyses them.

Adversary	Primary countermeasure	Analysis
Majority faction	Bridging aggregation (Level A)	Section 3
Biased item passing review	Empirical DIF on latent classes (Level B)	Section 4
Cartel	Random assignment; sublinear discount	Section 6
Sybil, whitewashing	External uniqueness; non-rotatable pseudonyms	Assumption A1
Brigading, herding	Unpredictable assignment; blind commit-reveal	Assumptions A2 and A3
Long-con reviewer	Evaluator score on honeypots; asymmetric updates; cap	Section 5
Dishonest signer	Deterministic recomputation by a threshold consortium	Assumption A4
Sample poisoning	Distinct respondents; batch validation	Section 8
Statistical deanonymization	Out of scope here (text normalization, delays)	Section 8

Assumption A4 (Reproducible computation). The scores are recomputed from the recorded inputs by a threshold of independent signers, and a result is accepted only if enough signers agree.

Assumption A5 (Respondent sampling). The respondents of a pilot are distinct persons. Their abilities and latent group memberships represent the population for which the item bank is intended.

In the reference implementation, Assumption A1 is realized with a threshold oblivious pseudorandom function over the state identity [13], threshold BBS+ blind credentials [1, 3, 49] and per-role nullifiers. Two parts are still missing: binding the OPRF input to the state-authenticated identity (roadmap item T20), and a distributed key generation that removes the trusted dealer from both committees (T19). Assumption A2 uses randomness derived from the latest signed checkpoint; the repository records that this derivation can still be ground by whoever orders the last deposits (roadmap item T37), so Assumption A2 is not yet met. Assumption A3 uses commitments that bind the committer and the item. Assumption A4 relies on a deterministic Rust engine (Section 7). Assumption A5 is the strongest assumption and the least under the system’s control, because respondents are self-selected members of the network (Section 8).

2.5 Lifecycle of an item

Figure 1 shows the lifecycle.

1. **Deposit.** An author deposits a draft item with a mandatory primary source.
2. **Admission.** Each epoch a lottery admits a subset of the deposits, because validation capacity, not proposals, is the bottleneck.
3. **Review.** Each admitted item is reviewed blind by $k = 7\text{--}11$ reviewers, assigned at random and stratified on their estimated latent position. Each reviewer declares a probability that the item will pass empirical validation; that number is both the rating used by Level A and the forecast scored by Level C.
4. **Bridging gate.** Level A computes a bridge score B_j . The item passes if $B_j \geq \tau + \varepsilon$. If B_j falls in the band $[\tau - \varepsilon, \tau + \varepsilon)$, the item gets more reviewers and is re-decided against τ .

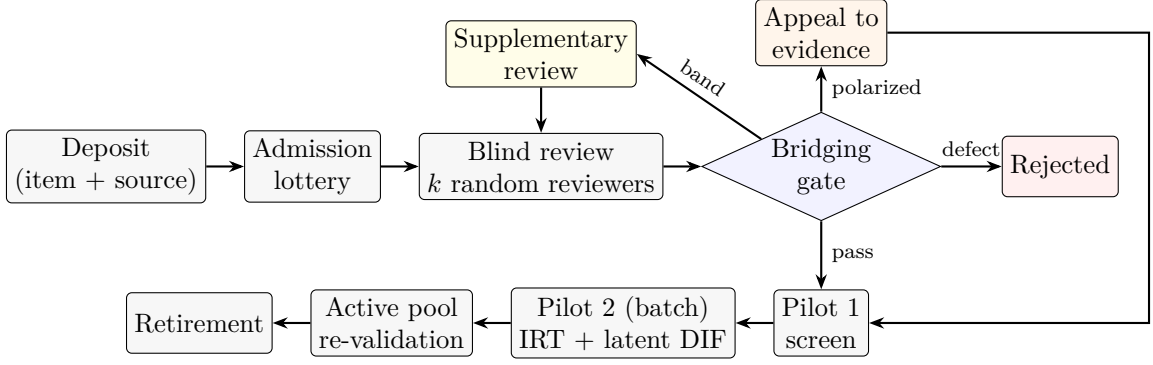


Figure 1: Lifecycle of an item. Level A (bridging) is only an upstream filter; the verdict comes from the pilots (Level B). A polarization rejection can be appealed directly to the pilots at the cost of the author’s reputation.

Below the band, it is rejected, or, if it was rejected for polarization (large $|f_j|$), its author may appeal to the evidence filter.

5. **Pilots.** Stage 1 (≈ 300 respondents) screens out broken and non-discriminating items. Stage 2 (batches, ≈ 3000 respondents for latent DIF) fits IRT and tests for DIF.
6. **Pool.** Surviving items enter the active pool. The pool is periodically re-validated as a whole, and items are retired for exposure, drift or emerging DIF.

3 Level A: bridging

3.1 Model and estimator

Let $\Omega \subseteq U \times J$ be the observed (reviewer, item) pairs of an epoch, with $|U| = n$ reviewers and $|J| = m$ items, and let $r_{uj} \in [0, 1]$ be the probability that reviewer u declares for item j . Each reviewer carries a weight $w_u \geq 0$ computed from the previous epoch: 0 during probation, 1 for the founders who bootstrap the network, and $\min(w_{\max}, E_u)$ afterwards, where E_u is the evaluator score of Section 5. The design then applies the anti-collusion discount of Section 6; the engine implements it, but the protocol does not yet apply it when it computes these weights. The model is the one-dimensional ($d = 1$) factorization

$$\hat{r}_{uj} = \mu + \iota_u + \iota_j + f_u f_j, \quad (1)$$

with a global mean μ , reviewer and item intercepts ι_u, ι_j (written b_u, b_j in the repository), a reviewer position f_u on a latent axis, and an item loading f_j . The parameters $\xi = (\mu, \iota_U, \iota_J, f_U, f_J)$ minimize

$$L(\xi) = \sum_{(u,j) \in \Omega} w_u (r_{uj} - \hat{r}_{uj})^2 + \lambda_b \left(\sum_u \iota_u^2 + \sum_j \iota_j^2 \right) + \lambda_f \left(\sum_u f_u^2 + \sum_j f_j^2 \right), \quad (2)$$

with $\lambda_b = 0.15$ and $\lambda_f = 0.03$, and μ unpenalized. The objective is non-convex. The reference implementation minimizes it with L-BFGS and a strong-Wolfe line search [27, 35] from eight seeded starts, keeps the lowest objective, and fixes the sign of the axis by making the largest $|f_j|$ non-negative.

The *bridge score* is pessimistic: $B_j = \min_s \iota_j^{(s)}$, taken over the full fit and 10 refits on subsamples that keep each observation with probability 0.85 [15]. The gate compares B_j with a threshold $\tau \approx 0.08$ and a band half-width $\varepsilon \approx 0.008$, both provisional (Section 2).

The design rationale, taken from Community Notes [50], is that heavy penalties on the intercepts force the model to explain approval through the factor term first. Approval that

a reviewer's side alone explains goes into $f_u f_j$. Only approval that no side explains, that is, approval *across* the divide, survives in ι_j . The rest of this section examines what the objective actually guarantees.

3.2 Invariances of the objective

The data term of (2) depends on ξ only through the predictions $\hat{r}_{uj}$. Several transformations leave every prediction unchanged:

$$\begin{aligned} \text{(G1)} \quad & \mu \mapsto \mu + c, \quad \iota_j \mapsto \iota_j - c \quad \forall j; & \text{(G2)} \quad & \mu \mapsto \mu + c, \quad \iota_u \mapsto \iota_u - c \quad \forall u; \\ \text{(G3)} \quad & f_u \mapsto f_u + c \quad \forall u, \quad \iota_j \mapsto \iota_j - c f_j \quad \forall j; & \text{(G4)} \quad & f_j \mapsto f_j + c \quad \forall j, \quad \iota_u \mapsto \iota_u - c f_u \quad \forall u; \\ \text{(G5)} \quad & f_u \mapsto s f_u, \quad f_j \mapsto f_j / s, \quad s \neq 0. \end{aligned}$$

(G3) translates the reviewers' axis and (G4) the items' axis; (G5) rescales the axis and, for $s = -1$, flips its sign. Along these directions only the penalties vary, so the penalties alone decide where the solution sits. This yields exact identities.

Lemma 1 (Stationarity identities). [proved] *For any weights $w_u \geq 0$ and any stationary point of (2) (in particular any local minimum),*

$$\sum_j \iota_j = 0 \quad \text{and} \quad \sum_u \iota_u = 0, \tag{3}$$

$$\lambda_b \sum_j f_j \iota_j = \lambda_f \sum_u f_u, \tag{4}$$

$$\lambda_b \sum_u f_u \iota_u = \lambda_f \sum_j f_j, \tag{5}$$

$$\sum_u f_u^2 = \sum_j f_j^2. \tag{6}$$

Proof. Let ξ^* be stationary and $\xi(c)$ a curve through it along one of the families above. The data term is constant along the curve, so $0 = \frac{d}{dc} L(\xi(c))|_{c=0}$ is the derivative of the penalty alone. For (G1) this derivative is $-2\lambda_b \sum_j \iota_j$, and (G2) is symmetric. For (G3) it is $\frac{d}{dc} [\lambda_b \sum_j (\iota_j - c f_j)^2 + \lambda_f \sum_u (f_u + c)^2]_{c=0} = -2\lambda_b \sum_j f_j \iota_j + 2\lambda_f \sum_u f_u$, and (G4) is symmetric. For (G5) it is $\frac{d}{ds} [\lambda_f (s^2 \sum_u f_u^2 + s^{-2} \sum_j f_j^2)]_{s=1} = 2\lambda_f (\sum_u f_u^2 - \sum_j f_j^2)$. $\square$

On the repository's reference dataset (200 reviewers, 10 items, a 60/40 split between two camps; `sim/bridging_irt_dif.py`), the best of eight fits has $\|\nabla L\|_\infty = 7 \times 10^{-7}$. Every identity holds to that precision: $\sum_j \iota_j = 1.9 \times 10^{-8}$, $\sum_u \iota_u = 4.2 \times 10^{-6}$, $\lambda_b \sum_j f_j \iota_j = \lambda_f \sum_u f_u = -3.899 \times 10^{-3}$, $\lambda_b \sum_u f_u \iota_u = \lambda_f \sum_j f_j = 6.426 \times 10^{-2}$ and $\sum_u f_u^2 = \sum_j f_j^2 = 8.568$.

Community Notes also penalizes μ [50]. Identity (3) then becomes $\sum_j \iota_j = \mu$ (with a common penalty weight), a constraint that is negligible when hundreds of thousands of notes are fitted jointly. Isegoria leaves μ unpenalized and fits one epoch's batch of tens to hundreds of items, so the identity binds.

3.3 The gate is relative to the batch

Proposition 2 (Batch relativity). [proved] *Let every term of the minimum defining B_j be a stationary point of (2). Then $\sum_j B_j \leq 0$. Hence the mean bridge score of every batch is non-positive, and if $\tau - \varepsilon > 0$, every batch contains an item below the band: no batch passes the gate in full.*

Proof. Fix one fit s_0 among those in the minimum. Then $B_j \leq \iota_j^{(s_0)}$ for every j , and $\sum_j \iota_j^{(s_0)} = 0$ by Lemma 1. If every $B_j \geq \tau - \varepsilon > 0$, then $\sum_j B_j > 0$, a contradiction. $\square$

Table 2: Bridge scores B_j of the six consensus items of the repository simulation, fitted inside the original ten-item batch and fitted alone (same ratings, same estimator). Threshold $\tau = 0.08$. Source: `scripts/levelA_relativity.py`.

Item	01	02	04	05	06	07
In the ten-item batch	+0.104	+0.081	+0.085	+0.101	+0.077	+0.082
Alone	+0.014	-0.017	-0.001	+0.011	-0.022	-0.014
Passes in batch / alone	yes/no	yes/no	yes/no	yes/no	no/no	yes/no

The proposition says that the gate does not test “is this item approved across the divide?”. It tests “is this item approved across the divide at least τ more than the average item fitted with it?”. Table 2 shows the effect on the repository’s dataset. We score six consensus items twice: inside the original ten-item batch, which also contains a true but divisive item, two partisan items and one mildly partisan item, and alone.

Empirical finding 3. [empirical] Five of the six consensus items pass when the batch contains divisive items and none passes when they are scored together without them. Their approval did not change; the reference point did.

Three consequences follow. First, pass rates depend on the composition of the epoch. A threshold “calibrated on the pilot”, as the design prescribes, is calibrated to the pilot’s mix of items. Second, the design documents found that the reference value of Community Notes (0.40) was unusable and that $\tau \approx 0.08$ worked, and that scores “cluster near the threshold”. Both observations are consistent with batch-centred scores. Third, whoever influences the composition of a batch influences who passes. For example, weak “decoy” items lower the batch average and raise the relative score of a target. The admission lottery and the proposal rate limit bound this lever but do not remove it.

Three corrections are possible, and we have not evaluated any of them [open]. One is to penalize μ and fit many epochs jointly, which dilutes the constraint. Another is to anchor the scale with reference items of known standing and express τ relative to them; the golden items that already enter every review queue (Section 5) could serve. The third is to declare the gate relative on purpose, as an explicit quota. The second fits the rest of the design best.

3.4 The origin of the axis decides how majoritarian the score is

The translation (G3) moves the origin of the reviewer axis and trades item intercept against loading. This matters because the position of the origin decides what “approval not explained by the axis” means. We make this precise in a two-camp model.

Consider camps A and B with shares π_A, π_B and imbalance $\kappa = \pi_B - \pi_A \geq 0$ (so B is the majority). For item j let m_{Aj}, m_{Bj} be the camps’ mean ratings, $\bar{m}_j = \frac{1}{2}(m_{Aj} + m_{Bj})$ the *camp-balanced* mean and $\Delta_j = \frac{1}{2}(m_{Bj} - m_{Aj})$ the item’s lean. The *camp-size-weighted* mean, which is what a plain average of all ratings estimates, is $\pi_A m_{Aj} + \pi_B m_{Bj} = \bar{m}_j + \kappa \Delta_j$. We say a stationary point has the *two-camp form* if:

- (T1) every reviewer of camp s has $f_u = \varphi_s$ and $\iota_u = 0$, with half-distance $h = \frac{1}{2}(\varphi_B - \varphi_A) > 0$ and midpoint $c = \frac{1}{2}(\varphi_A + \varphi_B)$; and
- (T2) the fit reproduces the camp means: $\mu + \iota_j + \varphi_s f_j = m_{sj}$ for $s \in \{A, B\}$ and every j .

Proposition 4 (Majoritarian leak). [proved] *Let a stationary point of (2) have the two-camp form. Let $S = \sum_j \Delta_j^2$ and $S_{\Delta m} = \sum_j \Delta_j(\bar{m}_j - \mu)$. Then:*

- (i) $f_j = \Delta_j/h$ and $\iota_j = \bar{m}_j - \mu - t \Delta_j$, where $t = c/h$ is the offset of the origin from the midpoint between the camps;

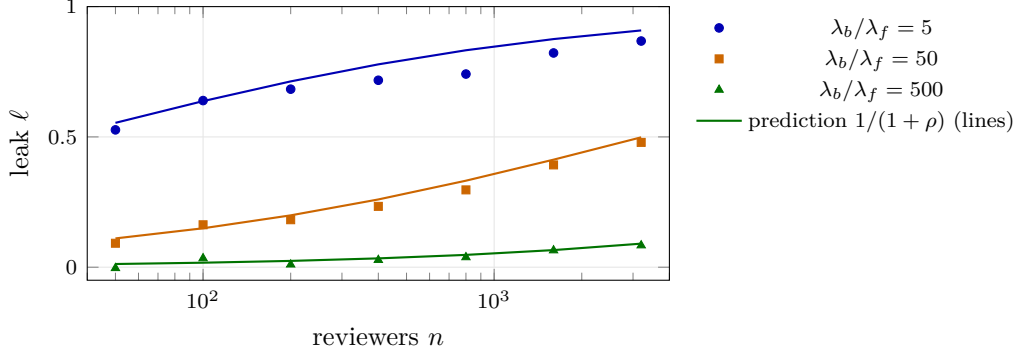


Figure 2: Majoritarian leak against the number of reviewers (points: full model; lines: $1/(1 + \rho)$ from Proposition 4). Data as in Table 3.

$$(ii) \quad t = \frac{\lambda_b S_{\Delta m} - \lambda_f n h^2 \kappa}{\lambda_b S + \lambda_f n h^2};$$

(iii) if $S_{\Delta m} = 0$, then

$$\mu + \iota_j = (1 - \ell) \bar{m}_j + \ell (\pi_A m_{Aj} + \pi_B m_{Bj}), \quad \ell = \frac{1}{1 + \rho}, \quad \rho = \frac{\lambda_b S}{\lambda_f n h^2}; \quad (7)$$

(iv) and, in that case, $\rho = \frac{\lambda_b}{\lambda_f} \gamma \sqrt{S/n}$ for some $\gamma \in [\sqrt{1 - \kappa^2}, 1]$.

Proof. Part (i) follows from subtracting and averaging the two equations of (T2). For (ii), substitute (i) and $\sum_u f_u = n_A \varphi_A + n_B \varphi_B = n h(t + \kappa)$ into (4), which gives $\lambda_b(S_{\Delta m} - tS)/h = \lambda_f n h(t + \kappa)$, and solve for t . Part (iii) is (i) with $t = -\kappa/(1 + \rho)$ from (ii). For (iv), (6) gives $n h^2(1 + t^2 + 2\kappa t) = S/h^2$. On $t \in [-\kappa, 0]$ the factor $1 + t^2 + 2\kappa t$ lies in $[1 - \kappa^2, 1]$, so $h^2 = \sqrt{S/(n(1 + t^2 + 2\kappa t))}$, and substituting into the definition of ρ gives (iv). $\square$

The *leak* ℓ is the fraction of the majority's weight that passes into the bridge score. At $\ell = 0$ the score is camp-balanced: each camp counts once, whatever its size. This is the property that motivates bridging. At $\ell = 1$ the score is the size-weighted mean, which is exactly what a plain average gives. The asymmetric penalty pushes ℓ toward 0, but $\lambda_b \gg \lambda_f$ is not the relevant condition. The relevant quantity is $\rho \approx (\lambda_b/\lambda_f)\sqrt{S/n}$, which compares how stiffly the penalty holds the item intercepts (summed over the batch's leans) with how stiffly it holds the reviewer positions (summed over the reviewers). At a fixed ratio λ_b/λ_f , the leak grows with the number of reviewers and falls only with the square root of the batch's total lean.

Assumptions (T1)–(T2) are idealizations. With regularization no fit is exact, and reviewers of a camp do not share one position. Table 3 therefore tests (7) on the full model: all parameters free, rating noise and reviewer severity as in the repository simulation, within-camp spread of positions, and nine ratings per reviewer. The design uses mirror pairs: ten consensus items and five pairs of partisan items with identical quality and opposite lean ($\Delta = \pm 0.36$), camps of 60% and 40% ($\kappa = 0.2$). The measured leak is $(\iota_{P+} - \iota_{P-})/(2\kappa\Delta)$, which is 0 for a camp-balanced score and 1 for a size-weighted one.

Empirical finding 5. [empirical] At the default penalties most of the camp-size effect survives in the bridge score: the leak is 0.53 with 50 reviewers and rises to 0.87 with 3,200. The closed form predicts every configuration within 0.1 and overestimates the leak in most of them; within-camp spread, absent from (T1), adds to $\sum_u f_u^2$ and hence to ρ . Raising λ_b/λ_f to 500 keeps the leak below 0.1 up to 3,200 reviewers.

Table 3: Majoritarian leak ℓ of the bridge score: measured on the full model (mean of three datasets; standard deviation in parentheses) against the prediction $1/(1+\rho)$ of Proposition 4 with $\gamma = 1$. Mirror-pair design, 60/40 camps. The default penalties are $\lambda_b/\lambda_f = 5$. Source: `scripts/levelA_leak.py`.

λ_b/λ_f	reviewers n	measured ℓ	(sd)	predicted $1/(1+\rho)$
5	50	0.53	(0.01)	0.55
5	100	0.64	(0.05)	0.64
5	200	0.68	(0.05)	0.71
5	400	0.72	(0.03)	0.78
5	800	0.74	(0.01)	0.83
5	1,600	0.82	(0.03)	0.88
5	3,200	0.87	(0.01)	0.91
50	50	0.09	(0.01)	0.11
50	100	0.16	(0.01)	0.15
50	200	0.18	(0.02)	0.20
50	400	0.23	(0.02)	0.26
50	800	0.30	(0.01)	0.33
50	1,600	0.39	(0.01)	0.41
50	3,200	0.48	(0.01)	0.50
500	50	0.00	(0.02)	0.01
500	100	0.03	(0.00)	0.02
500	200	0.01	(0.02)	0.02
500	400	0.03	(0.01)	0.03
500	800	0.04	(0.01)	0.05
500	1,600	0.06	(0.00)	0.07
500	3,200	0.08	(0.01)	0.09

Proposition 4 also explains an asymmetry in the repository simulation. There $n = 200$, $S \approx 0.39$ and $\kappa = 0.2$, so $\rho \approx 0.22$ and $\ell \approx 0.82$. The majority’s partisan item and the minority’s mirror-image item have the same quality and opposite lean. In the full fit they score -0.155 and -0.260 . The proposition predicts a gap of $\ell\kappa(\Delta_{08} - \Delta_{09}) \approx 0.82 \times 0.2 \times 0.72 \approx 0.12$; the measured gap is 0.105. Both items fail here. For items closer to τ , however, which of two mirror images passes depends on which camp is larger, which is what bridging is meant to prevent.

The normative content of bridging, “each side counts once”, therefore sits in a gauge. The objective reaches it only in the limit $\rho \rightarrow \infty$, and with more than two groups or a continuum of positions the “midpoint” is itself a choice. We see three remedies [open]:

- (a) scale the penalty ratio with $\sqrt{n/S}$; for example, $\rho \geq 9$ keeps $\ell \leq 0.1$;
- (b) fix the gauge explicitly, for instance by imposing $\sum_j f_j \iota_j = 0$, the $\rho \rightarrow \infty$ limit of (4). The origin then depends on $S_{\Delta m}$, the correlation between lean and approval within the batch, which is a new lever for anyone who controls the batch’s composition;
- (c) replace the intercept by a gauge-invariant statistic of the predictions, such as the smaller of the mean predicted ratings $\hat{r}_{uj}$ among reviewers in the lowest and in the highest decile of f_u . Predictions are invariant under (G1)–(G5), so any function of them is too, and a minimum across the spectrum does not depend on camp sizes.

3.5 What ratings cannot distinguish

Proposition 6 (Observational equivalence). [proved] *Let D be any decision rule, deterministic or randomized, that depends on the data only through the assignment Ω and the ratings (r_{uj}) . If two items induce the same distribution of ratings from every reviewer, then D has the same distribution of outcomes on both.*

Proof. The distribution of D 's output is a function of the joint distribution of its inputs, which is the same for both items. $\square$

The proposition is elementary, but it settles the most serious limitation of Level A. A true fact that one camp rejects and a false claim that the other camp accepts can polarize the same camps in the same way. When they do, no function of the ratings separates them, and every threshold trades lost true items for admitted partisan ones. The repository simulation shows this: at every τ in $[0.02, 0.12]$ the true but divisive item fails, while no partisan item passes. The remedy has to bring in information the ratings do not contain. That is the role of the appeal to evidence, which sends a polarization rejection to the pilots at the author's risk. Section 4.7 discusses how far that remedy reaches.

3.6 Capture cost in the reference simulation

In the repository simulation, 40 members of the majority camp rate the majority's partisan item at 1, and members of the opposing camp are added one group at a time. The bridge score moves from -0.14 (no opposing raters) through -0.10 , -0.06 , -0.02 , $+0.02$ and $+0.07$ to $+0.13$ (10, 20, 30, 40, 55 and 70 opposing raters). The crossing of $\tau = 0.08$ therefore lies between 55 and 70 of the 80 opposing reviewers (69–88%); linear interpolation gives about 57 (71%). The design documents quote “70 of 80 (87%)”, which is the upper end of this bracket [empirical]. This is one dataset and one seed. A capture-cost curve with confidence bands is part of the evaluation plan (Section 9). Proposition 4 adds that votes from the majority's own camp are not neutralized: they act through the leak.

3.7 Further identifiability conditions

The sign of the axis is fixed as described above. If the bipartite graph of Ω is disconnected, each component has its own gauge and the scores of different components are not comparable. Random assignment with $k \geq 7$ reviewers per item makes this unlikely, but the engine does not check it. The specification states that reviewers with fewer than $n_{\min} = 30$ ratings should not define the axis. This rule and the two-dimensional model are not implemented; the second is deferred by design decision D31 [29].

4 Level B: evidence

Level B decides. Items that pass the gate, or that are appealed, are trialled on respondents who answer them mixed with validated items. The verdict comes from the responses alone.

4.1 Ability and item screening

Respondent i 's ability is estimated as $\theta_i = (T_i - \bar{T})/\text{sd}(T)$, the standardized total score over a set of anchor items already certified free of DIF (30 in the repository simulations). For each trial item, a two-parameter logistic model (2PL) [2, 28]

$$P(X_{ij} = 1 \mid \theta_i) = \sigma(a_j(\theta_i - b_j)), \quad (8)$$

is fitted by logistic regression on the fixed θ , giving a discrimination a_j and a difficulty b_j . A fit that does not converge, for example under complete separation, is treated as undetermined. Stage 1 of the pilot keeps an item if its point-biserial correlation with θ is at least 0.20, its fit converged, and $a_j \geq 0.6$. A negative point-biserial reveals an inverted answer key, and the simulations catch one. Two gaps are acknowledged in the repository and remain open [open].

First, θ is a proxy on the standardized-total metric, while the thresholds come from a literature that uses an IRT metric [21]. Second, no guessing parameter is fitted, although multiple-choice items with four options are exactly where guessing matters (design decision D25).

4.2 DIF with observed groups: calibration only

With an observed group $g_i \in \{-1, +1\}$, DIF is tested by the logistic model [48] $\text{logit } P(X_{ij} = 1) = \beta_0 + \beta_1\theta_i + \beta_2g_i + \beta_3\theta_i g_i$, rejecting at $|\beta_2| > 0.40$, and by the Mantel–Haenszel common odds ratio with the ETS classes [22, 30, 51]. Both need a per-respondent group label. Requirement R1 forbids that label in production, so these tests are compiled only into closed calibration pilots with declared attributes (design decision D20). Production uses the latent variant below.

4.3 DIF on latent classes

The anonymity-compatible detector is a mixture IRT model [10, 40]. Each respondent belongs to an unobserved class $Z_i \in \{1, \dots, G\}$ with proportions π_g , and responses are independent given (θ_i, Z_i) :

$$P(X_{ij} = 1 \mid \theta_i, Z_i = g) = \sigma(a_{jg}(\theta_i - b_{jg})), \quad \ell(\cdot) = \sum_i \log \sum_g \pi_g \prod_j P(x_{ij} \mid \theta_i, g). \quad (9)$$

The implementation tries $G \in \{1, \dots, 4\}$, each with a shared discrimination ($a_{jg} = a_j$, uniform DIF) and with a per-class one. Every candidate is fitted from seeded starts with an analytic gradient, and the candidate with the lowest BIC [42] is kept. The search stops once an extra class no longer lowers the BIC. The item statistic is $\text{DIF}_j = \max_{g,h} |b_{jg} - b_{jh}|$ over classes holding at least 5% of the respondents. An item is flagged if the selected fit converged, has at least two classes, and $\text{DIF}_j > 1.0$. The literature value for this gap is 0.5. The repository reports that 0.5 flags every item of a batch with one biased item, keeps 1.0 as provisional, and defers the choice to a planned false-positive study. Batches must hold at least two items and 3,000 respondents. Mixture likelihood ratios are not χ^2 -distributed under the null [31, 43], so the BIC comparison is a heuristic, not a calibrated test.

4.4 Identification: why a single biased item is invisible

The repository simulation found that one distorted item in a batch of eight is invisible and two are detected. The following two propositions explain this.

Proposition 7 (One class-dependent item). [proved] *Let θ be observed without error and Z independent of θ , and suppose only item j 's response function depends on Z . Then the likelihood factorizes as $\prod_{k \neq j} P_k(x_k \mid \theta) \cdot [\sum_g \pi_g P_{jg}(x_j \mid \theta)]$, so the class parameters are identified only through the marginal response curve $m_j(\theta) = \sum_g \pi_g \sigma(a_{jg}(\theta - b_{jg}))$. For a symmetric two-class shift ($\pi = \frac{1}{2}$, $b_{j\pm} = b_j \pm \delta$, common a), with $x = a(\theta - b_j)$,*

$$m_j(\theta) = \sigma(x) + \frac{1}{2}(a\delta)^2 \sigma''(x) + O(\delta^4) = \sigma((1 - \frac{1}{4}(a\delta)^2)x) + \frac{1}{2}(a\delta)^2 \sigma'(x)(\frac{x}{2} - \tanh \frac{x}{2}) + O(\delta^4). \quad (10)$$

Proof. With only item j depending on Z , summing over Z acts on item j 's factor alone. The expansion averages $\sigma(x \pm a\delta)$, so odd orders cancel. It then uses $\sigma''(x) = -\sigma'(x) \tanh(x/2)$ and $\sigma((1 + \eta)x) = \sigma(x) + \eta x \sigma'(x) + O(\eta^2)$ with $\eta = -\frac{1}{4}(a\delta)^2$. $\square$

In words, a single class-dependent item looks, to second order, like a 2PL item with a lower discrimination. The residual $\frac{x}{2} - \tanh \frac{x}{2} = x^3/24 + O(x^5)$ vanishes to third order where the item is most informative. The same flattening is produced by error in θ , so the two are confounded in practice.

Table 4: Expected likelihood-ratio contribution at $N = 3000$ of one biased item (the part no 2PL absorbs) and of one biased pair (conditional mutual information), for $a = 1.25$, $b = 0$, $\pi = \frac{1}{2}$ and exact θ . For comparison, the BIC penalty per extra parameter is $\ln 3000 \approx 8$. Source: `scripts/levelB_information.py`.

shift δ	one biased item	one biased pair	ratio
0.15	0.000	0.15	306
0.30	0.008	2.25	296
0.45	0.038	10.72	283
0.60	0.117	31.32	269
0.90	0.532	130.17	245

Proposition 8 (Pairs of class-dependent items). [proved] *Under conditional independence given (θ, Z) ,*

$$\text{Cov}(X_j, X_k | \theta) = \sum_g \pi_g (P_{jg}(\theta) - m_j(\theta))(P_{kg}(\theta) - m_k(\theta)), \quad (11)$$

which for two classes equals $\pi(1 - \pi) \Delta_j(\theta) \Delta_k(\theta)$ with $\Delta_j = P_{j2} - P_{j1}$. Any one-class model implies $\text{Cov}(X_j, X_k | \theta) = 0$. Hence two items whose class-dependence overlaps in θ leave a signature that no one-class model absorbs.

Proof. $\mathbb{E}[X_j X_k | \theta] = \sum_g \pi_g P_{jg} P_{kg}$ and $\mathbb{E}[X_j | \theta] = m_j$. Expanding the covariance gives (11). For two classes, substitute $m_j = (1 - \pi)P_{j1} + \pi P_{j2}$. $\square$

Both signals are of fourth order in δ for small δ , but their constants differ greatly. Table 4 gives the expected likelihood-ratio contribution ($2N \times$ the per-respondent information) at $N = 3000$. For one item, this is the minimal expected Kullback–Leibler divergence between the mixture curve and any 2PL. For a pair, it is the conditional mutual information $I(X_j; X_k | \theta)$.

Empirical finding 9. [empirical] A biased pair carries 245–306 times the information of a single biased item. At $\delta = 0.9$ and $N = 3000$, a single item contributes 0.5 to the expected likelihood ratio, far below any penalty, while a pair contributes 130. The repository simulation, which uses a proxy θ , measures about 100 per biased pair. For small shifts, pairwise information scales as $\approx 0.048 \delta^4$ per respondent (the constant falls to 0.033 at $\delta = 0.9$, where the classes start to separate), so halving the shift requires about sixteen times the respondents for the same power. The production detector agrees. With exact θ it selects no mixture in 8 null batches and detects nothing in 8 batches with one biased item ($N \in \{3000, 6000\}$), and it flags exactly the biased items in all 8 batches with two, with no false flag.

Propositions 7 and 8 turn Requirement R7 from an empirical rule into a structural one. The latent classes are identified by *covariation between items* given ability, so they need at least two items that share a distortion. A campaign to tilt the bank leaves this signature. An isolated distortion does not, and neither does a single biased item repeated across batches that never contain another item biased the same way.

4.5 A confound: error in the ability proxy

The detector conditions on $\hat{\theta}$, a standardized anchor total, not on θ .

Proposition 10 (Proxy-induced dependence). [proved] *Let $\hat{\theta}$ be computed from anchor responses that are independent of the trial items given θ , and let every trial item’s response function be non-decreasing in θ . Then, with no DIF at all,*

$$\text{Cov}(X_j, X_k | \hat{\theta}) = \text{Cov}(P_j(\theta), P_k(\theta) | \hat{\theta}) \geq 0, \quad (12)$$

with strict inequality whenever $\text{Var}(\theta | \hat{\theta}) > 0$ and both functions are strictly increasing.

Table 5: Likelihood ratio of the two-class uniform-DIF mixture against one class, with θ from 30 anchors (proxy) or exact, for batches of 8 items of which 0, 1 or 2 are biased ($\delta = 0.9$). Mean of three datasets. Last row: BIC penalty $8 \ln N$ used by the repository simulation. Source: `scripts/levelB_proxy.py`.

θ	biased items	$N = 1500$	3000	6000	12000
proxy	0	53.7	91.2	169.0	281.9
proxy	1	48.2	84.5	152.7	239.3
proxy	2	67.1	219.7	433.4	568.8
exact	0	15.5	17.8	15.9	20.0
exact	1	13.6	12.1	18.9	18.2
exact	2	47.8	170.2	324.1	419.1
penalty $8 \ln N$		58.5	64.1	69.6	75.1

Table 6: Production detector (Rust engine and verdict rule) on null batches: eight clean items, $N = 6000$, four datasets per row, θ from the stated number of anchors. A flag here is a false positive. Source: `scripts/levelB_detector.py`.

anchors	mixture selected	largest clean gap	batches with a false flag	items flagged per batch
10	4/4	1.27–1.41	4/4	8 of 8
20	4/4	1.04–1.16	4/4	1–3
30	4/4	0.94–0.99	0/4	0
60	0/4	—	0/4	0

Proof. Conditional independence gives $\mathbb{E}[X_j X_k \mid \hat{\theta}] = \mathbb{E}[P_j(\theta) P_k(\theta) \mid \hat{\theta}]$ and $\mathbb{E}[X_j \mid \hat{\theta}] = \mathbb{E}[P_j(\theta) \mid \hat{\theta}]$. Two non-decreasing functions of the same random variable are non-negatively correlated (Chebyshev’s association inequality), and strictly so if both are strictly increasing and the variable is not degenerate. $\square$

Local dependence therefore exists whenever the proxy is imperfect, with or without DIF. A mixture whose classes shift all items in the same direction absorbs part of it, as a “higher than measured” and a “lower than measured” class. The likelihood ratio then grows linearly in N , while the BIC penalty grows like $\ln N$, so for large enough samples the BIC selects a mixture on data that contain none. The estimated gap of *every* item then equals the size of the common shift. This is the observed-score version of a problem known in DIF research: matching on an unreliable total inflates DIF, which is why SIBTEST corrects the matching variable by regression [45].

Table 5 shows the effect in a re-implementation of the repository simulation’s model. With exact θ the likelihood ratio stays flat when there is nothing to find, and also when there is one biased item, as Proposition 7 predicts. With the proxy it grows linearly in N even with zero biased items, and it exceeds the penalty from $N = 3000$ on. The mean estimated gap of clean items is then 0.71–0.86.

To measure the consequence for the production detector, we ran the Rust engine and the protocol’s verdict rule on null batches (no biased item, $N = 6000$) whose θ comes from 10, 20, 30 or 60 anchors (Table 6).

Empirical finding 11. [empirical] On null batches the production detector selects a spurious mixture whenever the proxy is imperfect enough. With 10 or 20 anchors it flags clean items in every batch. With 30 anchors, the configuration of the repository simulations, it flags none, but the largest clean gap reaches 0.94–0.99 against a threshold of 1.0. The provisional threshold is therefore calibrated, implicitly, to one anchor configuration. With the same 30 anchors and one biased item at $N = 6000$, the detector flags the biased item in 2 of 4 batches (gaps 1.15 and 1.03). It does so because the item rides on the artefactual class, not because its own shift is identified;

with exact θ it is never detected.

Three remedies are available [open].

- (a) Put the anchors inside the likelihood with class-invariant parameters and integrate θ out, which gives a full mixture IRT model. The anchors then identify class-specific ability distributions, and a class-wide shift is no longer mistaken for DIF. This is the principled fix. Its cost is computational.
- (b) Allow a class-specific ability shift and measure each item’s departure from the batch’s common class shift. In the two-class parametrization $b_{j\pm} = b_j \pm \delta_j$ of the repository simulation this is $2|\delta_j - \text{median}_k \delta_k|$. In our re-implementation this “differential gap” separates biased items (at least 1.65) from clean ones (at most 0.32) at both sample sizes, with 30 anchors (`scripts/levelB_differential.py`). A campaign that biases most of a batch in one direction would, however, be absorbed into the common shift, so this can only complement (a).
- (c) At a minimum, make the threshold a documented function of anchor reliability and calibrate it by simulation, as the evaluation plan requires.

4.6 Sample sizes and the minimum network

The pilot sizes of about 300 (screen) and 1500–3000 (DIF) are calibration targets taken from the psychometric literature, not derived results. The repository labels them a hypothesis [open]. Empirical finding 9 gives a first handle: for small shifts a biased pair contributes an expected likelihood ratio of about $2N \times 0.048 \delta^4$ for $a \approx 1.25$. Detection needs this to exceed the penalty of the extra class, which is roughly $(K+1) \ln N$ for a batch of K items. The respondent requirement sets a floor on the network’s size. A batch needs a few thousand *distinct* respondents in its validation window, and anonymity is weak in a small crowd. The repository estimates that the network cannot run as specified below about 2,000 active participants.

4.7 DIF is not bias

DIF shows that, at equal measured ability, an item’s difficulty depends on something else: the item measures a secondary dimension on which the classes differ [45]. Whether that secondary dimension is a nuisance is a judgment, not a statistical fact. For civic knowledge the judgment matters. Suppose one camp is systematically misinformed about a true fact. Then an item stating that fact shows DIF by construction, and Level B rejects it for the same reason Level A did. The appeal to evidence therefore recovers a true but divisive item only when the division lies in *opinions about the item*, and not when it lies in *knowledge of the fact*. The design currently treats every politically aligned secondary dimension as a nuisance. That is a normative choice, and we argue it should be made explicitly [design]. The same mechanism detects distortions on axes nobody chose to look for. In the repository simulation, an item neutral on the political axis shows DIF of 0.66 on education. Latent classes can surface such an axis, but only when a batch contains at least two items distorted along it.

5 Level C: reputation

Reputation enters in two separate scores on unlinkable pseudonyms (Requirement R4). The author score limits how much an author may propose. The evaluator score weights reviews.

5.1 Author score

Let $q_j \in [0, 1]$ be the final quality of author a 's item j , a function of its Level B statistics whose exact definition is still open. The author score is the shrunken, time-discounted average

$$C_a = \frac{\alpha_0 + \sum_j w_j q_j}{\alpha_0 + \beta_0 + \sum_j w_j}, \quad w_j = e^{-\Delta t_j/T}, \quad (13)$$

with prior $(\alpha_0, \beta_0) = (2, 3)$ and $T = 18$ months. Two accepted items out of two give $4/7 \approx 0.57$; 180 out of 200 give $182/205 \approx 0.89$. C_a sets the proposal rate $q_{\min} + (q_{\max} - q_{\min})C_a$. It never weights a vote [design]. A failed appeal to evidence is to be recorded as a negative pseudo-observation inside C_a , escrowed when the appeal is filed (design decision D27). The reference implementation still applies a simpler stake-and-refund rule.

5.2 Evaluator score: forecasts scored on golden items

Every reviewer declares a probability p_{uj} that item j will pass empirical validation. The same number is the rating r_{uj} of Level A. About 5% of every review queue consists of *golden* items. These are indistinguishable from the rest, and their outcome $o_j \in \{0, 1\}$ is known from validated history (design decision D21). The forecasts on golden items are scored. The design goal (design decision D6) is that honest forecasting is optimal, that copying the crowd earns nothing, and that being right where the crowd is wrong is what pays. Its opposite would be a Schelling-point mechanism that rewards agreement with the majority. The current rule is the Brier skill score against the crowd [5, 34]:

$$\text{BSS}_u = 1 - \frac{\sum_j (p_{uj} - o_j)^2}{\sum_j (\bar{p}_j - o_j)^2}, \quad \bar{p}_j = \frac{\sum_v w_v p_{vj}}{\sum_v w_v}, \quad (14)$$

where the crowd forecast $\bar{p}_j$ averages the whole panel, the reviewer included (design decision D23). The evaluator score is $E_u = \sigma(\gamma \text{BSS}_u)$. It is updated by an asymmetric moving average that rises slowly and falls fast, and it gives the weight $w_u = \min(w_{\max}, E_u)$ with $w_{\max} = 3 \text{ median}(w)$.

Scoring only golden items is more than a convenience. On a live item the outcome is observed only if the item reaches Level B, and whether it does depends on the forecasts themselves, because they are also the bridging ratings. When a forecast influences whether its outcome is observed, known results on decision scoring show that strict properness requires randomizing the decision [9, 36]. Golden items avoid the problem because their outcome is fixed in advance.

5.3 The ratio-form skill score is not proper

Proposition 12 (Non-properness of the skill score). [proved] *Score a single item with baseline $b \in (0, 1)$ that does not depend on the report, and let the reviewer believe $P(o = 1) = q \in (0, 1)$. The expected skill score*

$$\mathbb{E}[\text{BSS}] = 1 - \frac{q(1-p)^2}{(1-b)^2} - \frac{(1-q)p^2}{b^2}$$

is strictly concave in p , and its maximizer satisfies

$$\text{logit } p^* = \text{logit } q + 2 \text{logit } b. \quad (15)$$

Truthful reporting is optimal if and only if $b = \frac{1}{2}$.

Proof. The expectation over o is immediate. The first-order condition $q(1-p)/(1-b)^2 = (1-q)p/b^2$ gives $p/(1-p) = \frac{q}{1-q} \cdot \frac{b^2}{(1-b)^2}$, which is (15). The second derivative is $-2q/(1-b)^2 - 2(1-q)/b^2 < 0$. $\square$

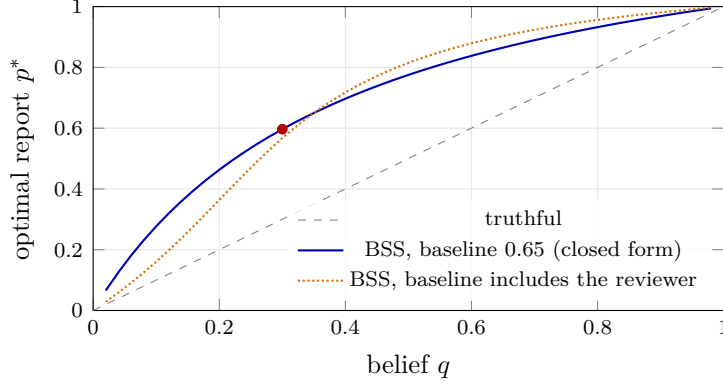


Figure 3: Report that maximizes the expected Brier skill score on one golden item, against the reviewer’s belief, when the other panelists’ mean forecast is 0.65. Solid line: baseline excluding the reviewer (Proposition 12). Dotted line: baseline including the reviewer with weight 1/9, as in the implementation (numerical). The marked point is a dissenter who believes 0.30 and is best off reporting 0.60. Source: `scripts/levelC_bss.py`.

Table 7: Distortion of the optimal report under the ratio-form skill score, $|p_j^* - q_j|$, for m jointly scored items with beliefs uniform on $[0.1, 0.9]$ and baselines uniform on $[0.2, 0.8]$ (200 random instances; 60 for $m = 8$). Source: `scripts/levelC_bss.py`.

m	baseline: others only		baseline includes reviewer	
	mean	max	mean	max
1	0.229	0.577	0.227	0.569
2	0.127	0.434	0.118	0.416
4	0.061	0.173	0.061	0.227
8	0.030	0.081	0.035	0.095

That skill scores built on a proper score are themselves improper, and that the resulting hedging shrinks as the number of scored forecasts grows, is known in forecast verification [33]. What matters here is the direction of the hedge. A ratio of two random sums is not an expectation of a score. The denominator gives more weight to outcomes where the crowd was right, and (15) shows the result: the report is pushed toward the outcome the crowd favours, by twice the crowd’s log-odds. With the crowd at 0.65, a reviewer who believes 0.78 should report 0.92, and a reviewer who believes 0.30 should report 0.60, crossing to the crowd’s side (Figure 3). The distortion is largest for the dissenter, who is exactly the reviewer the design wants to reward.

With several items the ratio couples them, and there is no closed form. Table 7 gives the exact optimum, enumerating all 2^m outcomes, for random beliefs and baselines.

Empirical finding 13. [empirical] The mean distortion falls roughly as $0.23/m$ in the number of jointly scored golden items, consistent with Murphy’s observation. Including the reviewer in the baseline, as the implementation does, changes it little. At a golden-item rate of 5%, a reviewer who judges about ten items per epoch meets a golden item about every other epoch. A score computed over a few golden items is therefore in the regime of the largest distortion.

5.4 A proper alternative

Proposition 14 (Leave-one-out difference score). [proved] Let $\bar{p}_{-u,j}$ be the weighted mean forecast of the other panelists on item j , and

$$S_u = \frac{1}{m} \sum_j \left[(\bar{p}_{-u,j} - o_j)^2 - (p_{u,j} - o_j)^2 \right]. \quad (16)$$

Under Assumption A3, so that $\bar{p}_{-u,j}$ does not depend on p_{uj} :

- (i) S_u is strictly proper: $\mathbb{E}[S_u] = c_u - \frac{1}{m} \sum_j (p_{uj} - q_j)^2$ with c_u not depending on the reports, so truthful reporting is the unique optimum;
- (ii) a reviewer who copies the crowd ($p_{uj} = \bar{p}_{-u,j}$) scores exactly 0 for every outcome, not only in expectation;
- (iii) $S_u > 0$ if and only if the reviewer's Brier score on the scored items beats the crowd's;
- (iv) S_u is additive over items, so it aggregates across epochs without reweighting.

Proof. For (i), use $\mathbb{E}[(p - o)^2] = (p - q)^2 + q(1 - q)$ for $o \sim \text{Bernoulli}(q)$ [20]. Parts (ii)–(iv) are immediate from (16). $\square$

The difference score keeps what D6 wants from the skill score, namely zero for copying the crowd and a positive score only for beating it, and drops the ratio that breaks properness. One new exposure deserves mention. Because the baseline is formed by the other panelists, a coalition can raise a member's score by degrading its own forecasts. This costs the saboteurs their own scores, is bounded by their share of the panel weight, and is blunted by the fact that golden items are indistinguishable from live ones, whose ratings also enter bridging.

5.5 From scores to weights

Proposition 15 (A cap that never binds). [proved] *If all weights lie in $(0, 1)$ and $\text{median}(w) \geq \frac{1}{3}$, then $w_{\max} = 3 \text{median}(w) \geq 1 > w_u$ for every u , so $\min(w_{\max}, E_u) = E_u$ always.*

With $E_u = \sigma(\gamma \text{BSS}_u)$ and a crowd baseline that puts the typical reviewer's score near zero, the median of E is close to $\frac{1}{2}$, so the cap is vacuous in the regime the design expects. The repository audit notes the same point. Expressing weights relative to the median, $w_u = E_u / \text{median}(E)$, or on the odds scale, $w_u = e^{\gamma S_u}$, would give the cap force [open].

A subtler issue concerns incentives. The weight is a non-linear function of the realized score, first through σ and then through a moving average that falls faster than it rises (a guard against the long con). A reviewer who maximizes expected future weight therefore faces a loss-averse objective, and reducing variance pays. Under Proposition 14 the zero-variance strategy is to copy the crowd. The asymmetry that deters the long con thus pulls in the same direction as herding, against D6. How strongly depends on γ , on the two rates and on the number of scored items, none of which the design specifies [open]. A natural mitigation is to accumulate the linear score (16) over a long window and map it to a weight only afterwards, so that each item's contribution stays small and truthful reporting remains approximately optimal.

5.6 Judgments without ground truth

Some dimensions never receive an empirical verdict, such as clarity or the quality of the cited source. For these the design proposes peer prediction. Candidates include the output-agreement score of Dasgupta and Ghosh [12], which rewards agreement with a peer on a shared item minus the agreement on unrelated items, multi-task correlated agreement [46], and the Bayesian truth serum [38]. Only the pairwise Dasgupta–Ghosh score is implemented. How it is aggregated over pairs and items, and the incentive analysis in the presence of reputation weights, remain open [open].

6 Coordination and collusion

An individual weight cap does not stop a cartel of many modest members. The design therefore discounts correlated reviewers as a group, in the spirit of quadratic funding [6].

6.1 The sublinear discount

Reviewers are clustered by the connected components of the graph that joins u and v when the correlation of their judgment vectors satisfies $|\rho_{uv}| \geq \rho_0$. A cluster G with raw weight $s_G = \sum_{u \in G} w_u$ contributes $\min(s_G, s_G^\alpha)$, with $\alpha = 0.5$, split among members in proportion to their raw weight. Each member's weight is therefore multiplied by $\min(1, s_G^{\alpha-1})$. The engine implements and tests this discount; the protocol does not yet apply it to the bridging weights.

Lemma 16. [proved] *The discount never increases a weight. A cluster of k members of unit weight counts k^α (500 coordinated reviewers count as $\sqrt{500} \approx 22$), and a singleton of weight at most 1 is unchanged.*

Proof. $\min(1, s^{\alpha-1}) \leq 1$. For $s = k \geq 1$ the factor is $k^{\alpha-1}$, and $k \cdot k^{\alpha-1} = k^\alpha$. For a singleton with $w \leq 1$, $s^{\alpha-1} \geq 1$, so the factor is 1. $\square$

The min matters. Without it a cluster with total weight below 1, which is every honest singleton with $E_u \in (0, 1)$, would be *boosted*; the repository audit found and fixed this defect.

6.2 The discount is only as strong as its detector

Proposition 17 (Evasion pays). [proved] *Let a cartel of total weight $s \geq 1$ be detected as j clusters of equal weight $s/j \geq 1$. Its total influence $j(s/j)^\alpha = j^{1-\alpha}s^\alpha$ is strictly increasing in j for $\alpha < 1$. It reaches the undiscounted s when every member escapes detection.*

The discount thus converts detection recall directly into protection, and every evasion of the detector pays. The repository audit reports that small independent noise ($\sigma = 0.05$) added to identical votes already defeats the correlation clustering. Two further weaknesses are known. Honest like-minded reviewers are discounted as if they were a cartel. Connected components chain, so intermediaries can merge unrelated groups, which invites griefing. A robust coordination statistic, with a false-positive rate stated for honest like-minded reviewers, is open [open].

6.3 Within an epoch there is almost nothing to correlate

Proposition 18 (Sparse overlap). [proved] *If each reviewer rates r of the m items of an epoch, chosen uniformly at random, then the number of items two given reviewers share is hypergeometric with mean r^2/m .*

With $r = 9$ the expected overlap is 1.62, 0.81 and 0.27 items per epoch for $m = 50, 100$ and 300. Accumulating 30 shared items, a modest basis for a correlation, takes about $30m/r^2$ epochs: 19, 37 and 111 respectively. The correlation matrix is therefore undefined within an epoch at design scale. The discount can act only on long histories, while a coalition can form and act within a single epoch. Per epoch, the operative defence is random assignment (Assumption A2). Table 8 gives the probability that a coalition holding a share s of the eligible reviewers wins a majority of a panel of nine drawn uniformly. Stratified assignment further confines a one-camp coalition to its camp's share of the seats. Under bridging, a one-camp majority is in any case not sufficient, although Proposition 4 shows that it helps. The threat that matters is a coalition that spans the divide, which is what the capture-cost experiment of Section 3 measures.

Table 8: Probability that a coalition holding a share s of the eligible reviewers holds at least 5 of the 9 seats of a uniformly drawn panel. Source: `scripts/assignment.py`.

share s	0.1	0.2	0.3	0.4	0.5
$P(\geq 5 \text{ of } 9)$	0.0009	0.020	0.099	0.267	0.500

7 Reference implementation and evidence

7.1 Architecture and determinism

The reference implementation [29] is a Rust workspace of four crates. The `scoring` crate holds Levels A, B and C and the anti-collusion discount. It has no dependency on the identity or network crates, so the compiler guarantees that the engine runs offline on recorded inputs. `identity` implements enrollment, credentials and nullifiers; `network` implements content addressing, Merkle trees, a transparency log [11, 26], threshold checkpoints, erasure coding [39] and anchoring; `protocol` orchestrates the lifecycle as a state machine that rejects invalid transitions.

Requirement R6 is enforced by construction. Observations are sorted into a canonical order before every fit. Every random draw comes from a seeded ChaCha8 stream with a fixed consumption order. No computation iterates over a hash map. Floating-point sorts use a total order. Reproducibility is tested bit for bit within a platform; cross-platform agreement has not yet been tested.

7.2 Oracles and tests

The Python simulations in `sim/` are the executable specification. Their outputs are exported as fixtures that the Rust engine must reproduce, and a drift check regenerates them under a pinned NumPy/SciPy environment. The engine departs from the simulations deliberately, and the repository documents where: eight starts with the lowest objective kept, a strong-Wolfe line search, and warm-started bootstrap refits whose minimum includes the full fit. Line coverage is about 97%, but coverage measures execution, not verification. For that reason the repository adds property-based tests (determinism, invariance to input order, zero weight equals absence, sign symmetry, analytic gradients against finite differences) and mutation testing. Of 1,482 mutants, the remaining survivors are 32 mutants justified as equivalent. The byte decoders are also fuzzed. None of this checks that the *design* is right. The analyses of Sections 3 to 6 are meant to fill that gap.

7.3 Evidence status

Table 9 merges the repository’s own audit with the results of this paper. No claim has been independently reviewed, and none is scientifically characterized in the sense of having false-positive and power tables with confidence intervals.

8 Limitations

8.1 Structural limits of the design

The following are properties of the design, not defects of the implementation.

- L1. **True but divisive items.** No rule that reads only ratings separates them from partisan items (Proposition 6). The appeal to evidence recovers them only when the division concerns opinions about the item and not knowledge of the fact (Section 4.7).

Table 9: Evidence status of the main claims. “Audit” is the repository’s claim/evidence matrix at the described commit; “This paper” is what the analysis above adds.

Claim	Audit	This paper
Bridging objective as specified ($d = 1$)	tested	identities of Lemma 1 hold to optimizer precision
Latent axis recovered ($ r = 0.99$)	tested, one seed	—
Polarized items rejected	tested, one dataset	score is batch-relative (Proposition 2) and partly majoritarian (Proposition 4)
Capture needs 87% of the other camp	tested qualitatively	crossing lies at 69–88%, $\approx 71\%$ interpolated
Reputation weights enter the fit	implemented	cap never binds (Proposition 15)
Mixture DIF detects ≥ 2 biased of 8	tested, reproduced	explained by Proposition 8
One biased item of 8 is invisible	tested	explained by Proposition 7
False-positive rate of latent DIF	not established	proxy-induced false flags (Empirical finding 11)
Pilot sizes 300 / 1500 / 3000	hypothesis	pair information $\propto N\delta^4$
Evaluator score rewards correct dissent	resolved (crowd baseline)	not proper (Proposition 12); proper fix (Proposition 14)
Identical cartel counts $\sqrt{k}$	tested	evasion pays (Proposition 17)
Jittered cartel detected	unsolved	per-epoch overlap r^2/m (Proposition 18)
Randomness cannot be ground	partial (T37)	Assumption A2 not yet met

- L2. **Bias of the pool, not of an item.** Every item can pass DIF while the choice of topics is skewed. The design constrains coverage with per-domain quotas set by stratified sortition [17].
- L3. **The elite blind spot.** An item can be neutral on the political axis yet distorted on a socio-economic one. Latent classes can find such an axis, but only with at least two items distorted along it in the same batch (Proposition 8).
- L4. **Competence is a chosen construct.** Psychometrics checks that a test measures one thing coherently, not that it is the right thing to measure.
- L5. **The initial consortium.** Whom to trust first is a human decision. Reproducible computation and the freedom to fork make a bad choice survivable, not impossible.
- L6. **Weighted voting is contested.** If a bank were used to weight votes by competence, the objection is political, not technical: even a neutral test weights participation by variables correlated with education and income [4, 16]. Certification, licensing and fact-checking are less contested uses of the same machinery.

8.2 Threats to the validity of this analysis

- **Stylized models.** Proposition 4 assumes two camps and an exact fit. Propositions 7 and 8 assume exact ability. We test each on the full model, but only with synthetic data generated with the repository’s own parameters and a few seeds per configuration.
- **Respondents are self-selected.** Assumption A5 is the least defensible assumption. The respondents are members of the network. If they differ from the target population, item parameters and DIF verdicts may not transfer. We have not analysed coordinated respondents who answer strategically (sample poisoning).
- **Scope.** Statistical deanonymization (stylometry, timing), the security of the identity layer and the behaviour of the network are outside this paper.

- **Self-review.** The analysis was carried out by the designer, with the assistance of an AI system, and has not been reviewed independently. The repository’s own methodology does not accept self-review as primary evidence. This paper is written in part to invite that review.

9 Evaluation plan

The findings above are single-regime results. We commit in advance to the following study before any threshold is declared final. It extends roadmap items T24 and T25 of the repository. All code, seeds and results will be versioned in the repository. We will report estimates with confidence intervals, never only point values.

Level A. *Factors:* number of reviewers (50–5,000); items per batch (10–300); camp split (50/50 to 90/10) and number of camps; rating noise; ratings per reviewer and reviewers per item; penalty ratio (5–500); and three gauge variants: implicit (current), explicit $\sum_j f_j \iota_j = 0$, and the gauge-invariant decile minimum. We will also vary batch composition: the share of divisive items and the presence of decoys. *Metrics:* the leak ℓ ; false negatives on consensus items; false positives on partisan items; capture-cost curves; stability across seeds. *Decision rule:* adopt the variant and penalties that keep $\ell \leq 0.1$ and the false-negative rate on consensus items at or below 5% across all batch compositions. If none does, report that bridging cannot meet both targets at that scale.

Level B. *Factors:* respondents (1,500–12,000); items per batch (4–16); number of biased items (0 to all); shift δ (0.2–1.2); class balance (0.1–0.5); uniform and non-uniform DIF; number and reliability of anchors (10–60); ability treatment (proxy, anchors in the likelihood, differential gap); one or two distorting axes. *Metrics:* item-level false-positive rate with *zero* biased items (the primary target); power surfaces; the threshold as a function of anchor reliability. *Decision rule:* choose the threshold that holds the item-level false-positive rate at or below 5% at the lowest anchor reliability the deployment will accept, with a multiplicity correction across the batch, and report the power at that threshold. The pilot sizes follow from the power tables, not from prose.

Level C. Simulated reviewer populations with heterogeneous skill and strategic types: truthful, herding, contrarian, and saboteur coalitions. We will compare the ratio score with the leave-one-out difference score, each under the logistic map and the asymmetric update. *Metrics:* drift of reports toward the crowd; rank correlation between E_u and true skill; time to detect a long-con reviewer.

Coordination. Jittered cartels, sparse histories, honest like-minded groups and chaining attacks. *Metrics:* the detection ROC, and the influence a coalition obtains within one epoch and over many epochs.

10 Related work

Bridging. Community Notes (formerly Birdwatch) scores crowd-written notes with the matrix factorization that Level A adopts, with a heavier penalty on intercepts than on factors. It reports that the notes selected this way are, on average, strongly informative [50]. Polis clusters opinion spaces in order to surface statements that find agreement across groups [47]. Ovadya and Thorburn [37] frame both as “bridging systems” and list open problems. Matrix factorization for

ratings is standard in recommender systems [25]. Our contribution here is analytic. We show that in the bridging objective the intercept is fixed only by the regularization, through a gauge, and we quantify how much of the majority’s weight leaks into it (Proposition 4). We also show that with an unpenalized mean the scores are relative to the batch (Proposition 2).

Psychometrics. The 2PL and 3PL models go back to Birnbaum [2]; see Lord [28] and Hambleton et al. [21]. DIF methods include Mantel–Haenszel with the ETS classification [22, 30, 51] and logistic regression [48]; Holland and Wainer [23] survey the field. SIBTEST treats DIF as multidimensionality and corrects the matching variable for measurement error [45], which is the observed-score counterpart of our Proposition 10. Mixture Rasch and mixture IRT models [40], and their use for detecting DIF without observed groups [10], are the basis of Level B. The irregularity of likelihood ratios for mixtures is classical [31, 43].

Elicitation and scoring. Proper scoring rules are surveyed by Gneiting and Raftery [20], after Brier [5] and Savage [41]; skill scores are discussed by Murphy [34]. When predictions influence which outcome is observed, strict propriety requires randomized decision rules [9, 36]; this is why Isegoria scores forecasts on golden items only. Without ground truth, peer-prediction mechanisms elicit honest reports [12, 32, 38, 46]. That skill scores are improper and encourage hedging was shown by Murphy [33]. Proposition 12 gives the hedge in closed form against a crowd baseline and shows that in this design it pushes dissenters toward the crowd, the reverse of the intended incentive.

Identity, reputation and collusion. The Sybil attack [14] and the social cost of cheap pseudonyms [18] motivate external uniqueness and non-rotatable pseudonyms. The identity layer builds on oblivious pseudorandom functions [13, 24], BBS-family signatures [1, 3, 7, 49], secret sharing and distributed key generation [19, 44], and equality-of-discrete-log proofs [8]. Sublinear group weights echo quadratic funding [6]. Stratified sortition for governance roles follows the literature on citizens’ assemblies [17].

11 Conclusion

The central idea of Isegoria is to use opinion as a filter and evidence as the verdict, so that the decision rests on what an item *does* rather than on what the network *thinks*. That idea survives the analysis. Each level, however, has a property that the design documents did not anticipate:

- bridging scores are relative to their batch, and at the default penalties they retain most of the camp-size effect that bridging is meant to remove;
- latent-class DIF can mistake error in the ability proxy for a latent class;
- the evaluator’s skill score is not proper, and the weight cap never binds;
- the anti-collusion discount depends on a detector that has almost no data within an epoch.

Each finding comes with a candidate correction. None of the corrections is validated yet. The next steps are the evaluation plan of Section 9, which will turn provisional thresholds into characterized ones, and independent review of both the statistics and the cryptography. The paper is versioned with the repository and will be revised as those results arrive.

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References

- [1] Man Ho Au, Willy Susilo, and Yi Mu. Constant-size dynamic k-TAA. In *Security and Cryptography for Networks (SCN 2006)*, volume 4116 of *Lecture Notes in Computer Science*, pages 111–125. Springer, 2006.
- [2] Allan Birnbaum. Some latent trait models and their use in inferring an examinee’s ability. In Frederic M. Lord and Melvin R. Novick, editors, *Statistical Theories of Mental Test Scores*. Addison-Wesley, Reading, MA, 1968.
- [3] Dan Boneh, Xavier Boyen, and Hovav Shacham. Short group signatures. In *Advances in Cryptology – CRYPTO 2004*, volume 3152 of *Lecture Notes in Computer Science*, pages 41–55. Springer, 2004.
- [4] Jason Brennan. *Against Democracy*. Princeton University Press, Princeton, NJ, 2016.
- [5] Glenn W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950.
- [6] Vitalik Buterin, Zoë Hitzig, and E. Glen Weyl. A flexible design for funding public goods. *Management Science*, 65(11):5171–5187, 2019.
- [7] Jan Camenisch and Anna Lysyanskaya. Signature schemes and anonymous credentials from bilinear maps. In *Advances in Cryptology – CRYPTO 2004*, volume 3152 of *Lecture Notes in Computer Science*, pages 56–72. Springer, 2004.
- [8] David Chaum and Torben P. Pedersen. Wallet databases with observers. In *Advances in Cryptology – CRYPTO ’92*, volume 740 of *Lecture Notes in Computer Science*, pages 89–105. Springer, 1993.
- [9] Yiling Chen, Ian A. Kash, Michael Ruberry, and Victor Shnayder. Eliciting predictions and recommendations for decision making. *ACM Transactions on Economics and Computation*, 2(2):6:1–6:27, 2014.
- [10] Allan S. Cohen and Daniel M. Bolt. A mixture model analysis of differential item functioning. *Journal of Educational Measurement*, 42(2):133–148, 2005.
- [11] Scott A. Crosby and Dan S. Wallach. Efficient data structures for tamper-evident logging. In *Proceedings of the 18th USENIX Security Symposium*, 2009.
- [12] Anirban Dasgupta and Arpita Ghosh. Crowdsourced judgement elicitation with endogenous proficiency. In *Proceedings of the 22nd International Conference on World Wide Web (WWW)*, pages 319–330, 2013.
- [13] Alex Davidson, Armando Faz-Hernández, Nick Sullivan, and Christopher A. Wood. Oblivious pseudorandom functions (OPRFs) using prime-order groups. RFC 9497, 2023.
- [14] John R. Douceur. The Sybil attack. In *Peer-to-Peer Systems (IPTPS 2002)*, volume 2429 of *Lecture Notes in Computer Science*, pages 251–260. Springer, 2002.

- [15] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall, New York, 1993.
- [16] David M. Estlund. *Democratic Authority: A Philosophical Framework*. Princeton University Press, Princeton, NJ, 2008.
- [17] Bailey Flanigan, Paul Gözl, Anupam Gupta, Brett Hennig, and Ariel D. Procaccia. Fair algorithms for selecting citizens’ assemblies. *Nature*, 596:548–552, 2021.
- [18] Eric J. Friedman and Paul Resnick. The social cost of cheap pseudonyms. *Journal of Economics & Management Strategy*, 10(2):173–199, 2001.
- [19] Rosario Gennaro, Stanislaw Jarecki, Hugo Krawczyk, and Tal Rabin. Secure distributed key generation for discrete-log based cryptosystems. *Journal of Cryptology*, 20(1):51–83, 2007.
- [20] Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.
- [21] Ronald K. Hambleton, Hariharan Swaminathan, and H. Jane Rogers. *Fundamentals of Item Response Theory*. Sage, Newbury Park, CA, 1991.
- [22] Paul W. Holland and Dorothy T. Thayer. Differential item performance and the Mantel–Haenszel procedure. In Howard Wainer and Henry I. Braun, editors, *Test Validity*, pages 129–145. Lawrence Erlbaum Associates, Hillsdale, NJ, 1988.
- [23] Paul W. Holland and Howard Wainer, editors. *Differential Item Functioning*. Lawrence Erlbaum Associates, Hillsdale, NJ, 1993.
- [24] Stanislaw Jarecki, Aggelos Kiayias, Hugo Krawczyk, and Jiayu Xu. TOPPSS: Cost-minimal password-protected secret sharing based on threshold OPRF. In *Applied Cryptography and Network Security (ACNS 2017)*, volume 10355 of *Lecture Notes in Computer Science*, pages 39–58. Springer, 2017.
- [25] Yehuda Koren, Robert Bell, and Chris Volinsky. Matrix factorization techniques for recommender systems. *Computer*, 42(8):30–37, 2009.
- [26] Ben Laurie, Adam Langley, and Emilia Kasper. Certificate transparency. RFC 6962, 2013.
- [27] Dong C. Liu and Jorge Nocedal. On the limited memory BFGS method for large scale optimization. *Mathematical Programming*, 45:503–528, 1989.
- [28] Frederic M. Lord. *Applications of Item Response Theory to Practical Testing Problems*. Lawrence Erlbaum Associates, Hillsdale, NJ, 1980.
- [29] Davide Macchia. Isegoria: a distributed network for producing and validating quiz questions (reference implementation and design documents). <https://github.com/DavideMacchia/isegoria>, 2026. Commit 2ee5e79. Design documents in `docs/`, simulations in `sim/`.
- [30] Nathan Mantel and William Haenszel. Statistical aspects of the analysis of data from retrospective studies of disease. *Journal of the National Cancer Institute*, 22(4):719–748, 1959.
- [31] Geoffrey McLachlan and David Peel. *Finite Mixture Models*. Wiley, New York, 2000.
- [32] Nolan Miller, Paul Resnick, and Richard Zeckhauser. Eliciting informative feedback: The peer-prediction method. *Management Science*, 51(9):1359–1373, 2005.

- [33] Allan H. Murphy. Hedging and skill scores for probability forecasts. *Journal of Applied Meteorology*, 12(1):215–223, 1973.
- [34] Allan H. Murphy. Skill scores based on the mean square error and their relationships to the correlation coefficient. *Monthly Weather Review*, 116(12):2417–2424, 1988.
- [35] Jorge Nocedal and Stephen J. Wright. *Numerical Optimization*. Springer, New York, 2nd edition, 2006.
- [36] Abraham Othman and Tuomas Sandholm. Decision rules and decision markets. In *Proceedings of the 9th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, pages 625–632, 2010.
- [37] Aviv Ovadya and Luke Thorburn. Bridging systems: Open problems for countering destructive divisiveness across ranking, recommenders, and governance. arXiv:2301.09976, 2023.
- [38] Dražen Prelec. A Bayesian truth serum for subjective data. *Science*, 306(5695):462–466, 2004.
- [39] Irving S. Reed and Gustave Solomon. Polynomial codes over certain finite fields. *Journal of the Society for Industrial and Applied Mathematics*, 8(2):300–304, 1960.
- [40] Jürgen Rost. Rasch models in latent classes: An integration of two approaches to item analysis. *Applied Psychological Measurement*, 14(3):271–282, 1990.
- [41] Leonard J. Savage. Elicitation of personal probabilities and expectations. *Journal of the American Statistical Association*, 66(336):783–801, 1971.
- [42] Gideon Schwarz. Estimating the dimension of a model. *The Annals of Statistics*, 6(2):461–464, 1978.
- [43] Steven G. Self and Kung-Yee Liang. Asymptotic properties of maximum likelihood estimators and likelihood ratio tests under nonstandard conditions. *Journal of the American Statistical Association*, 82(398):605–610, 1987.
- [44] Adi Shamir. How to share a secret. *Communications of the ACM*, 22(11):612–613, 1979.
- [45] Robin Shealy and William Stout. A model-based standardization approach that separates true bias/DIF from group ability differences and detects test bias/DTF as well as item bias/DIF. *Psychometrika*, 58(2):159–194, 1993.
- [46] Victor Shnayder, Arpit Agarwal, Rafael Frongillo, and David C. Parkes. Informed truthfulness in multi-task peer prediction. In *Proceedings of the 2016 ACM Conference on Economics and Computation (EC)*, 2016.
- [47] Christopher Small, Michael Bjorkegren, Timo Erkkilä, Lynette Shaw, and Colin Megill. Polis: Scaling deliberation by mapping high dimensional opinion spaces. *Recerca. Revista de Pensament i Anàlisi*, 26(2), 2021. doi: 10.6035/recerca.5516.
- [48] Hariharan Swaminathan and H. Jane Rogers. Detecting differential item functioning using logistic regression procedures. *Journal of Educational Measurement*, 27(4):361–370, 1990.
- [49] Stefano Tessaro and Chenzhi Zhu. Revisiting BBS signatures. In *Advances in Cryptology – EUROCRYPT 2023*, Lecture Notes in Computer Science. Springer, 2023.

- [50] Stefan Wojcik, Sophie Hilgard, Nick Judd, Delia Mocanu, Stephen Ragain, M. B. Fallin Hunzaker, Keith Coleman, and Jay Baxter. Birdwatch: Crowd wisdom and bridging algorithms can inform understanding and reduce the spread of misinformation. arXiv:2210.15723, 2022.
- [51] Michael Zieky. Practical questions in the use of DIF statistics in test development. In Paul W. Holland and Howard Wainer, editors, *Differential Item Functioning*, pages 337–347. Lawrence Erlbaum Associates, Hillsdale, NJ, 1993.

A Derivations

A.1 Proposition 4 step by step

Under (T1)–(T2), camp s satisfies $\mu + \iota_j + \varphi_s f_j = m_{sj}$. Subtracting the equation of camp A from that of camp B gives $(\varphi_B - \varphi_A)f_j = m_{Bj} - m_{Aj}$, that is, $2hf_j = 2\Delta_j$, so $f_j = \Delta_j/h$. Averaging the two equations gives $\mu + \iota_j + cf_j = \bar{m}_j$, so $\iota_j = \bar{m}_j - \mu - (c/h)\Delta_j$. This is part (i).

For (ii), the reviewer positions sum to $\sum_u f_u = n_A(c - h) + n_B(c + h) = nc + h(n_B - n_A) = nh(t + \kappa)$. The left side of (4) is $\lambda_b \sum_j f_j \iota_j = \frac{\lambda_b}{h} \sum_j \Delta_j (\bar{m}_j - \mu - t\Delta_j) = \frac{\lambda_b}{h} (S_{\Delta m} - tS)$. Equating the two and multiplying by h gives $\lambda_b(S_{\Delta m} - tS) = \lambda_f nh^2(t + \kappa)$, which is linear in t and solves to (ii).

For (iii), with $S_{\Delta m} = 0$ we get $t = -\kappa \lambda_f nh^2 / (\lambda_b S + \lambda_f nh^2) = -\kappa / (1 + \rho)$. Then $\mu + \iota_j = \bar{m}_j + \frac{\kappa}{1+\rho} \Delta_j = (1 - \ell)\bar{m}_j + \ell(\bar{m}_j + \kappa \Delta_j)$, and $\bar{m}_j + \kappa \Delta_j = \pi_A m_{Aj} + \pi_B m_{Bj}$ because $\pi_A + \pi_B = 1$.

For (iv), (6) with $f_u \in \{c - h, c + h\}$ gives $\sum_u f_u^2 = n(c^2 + h^2 + 2\kappa ch) = nh^2(1 + t^2 + 2\kappa t)$ and $\sum_j f_j^2 = S/h^2$, hence $h^4 = S/(n(1 + t^2 + 2\kappa t))$. The function $g(t) = 1 + t^2 + 2\kappa t$ is convex with its minimum $1 - \kappa^2$ at $t = -\kappa$ and $g(0) = 1$, so $g \in [1 - \kappa^2, 1]$ on $[-\kappa, 0]$, which contains t by (iii). Substituting $h^2 = \sqrt{S/(ng(t))}$ into $\rho = \lambda_b S / (\lambda_f nh^2)$ gives $\rho = (\lambda_b / \lambda_f) \sqrt{g(t)} \sqrt{S/n}$, so $\gamma = \sqrt{g(t)} \in [\sqrt{1 - \kappa^2}, 1]$. Since t depends on ρ , this is a fixed-point characterization. The bounds on γ make the explicit approximation $\rho \approx (\lambda_b / \lambda_f) \sqrt{S/n}$ accurate to a factor $\sqrt{1 - \kappa^2}$, which is 0.98 for $\kappa = 0.2$.

A.2 Information computations for Table 4

Ability is integrated against $\mathcal{N}(0, 1)$ with an 80-node Gauss–Hermite rule. For one biased item, the per-respondent information that no 2PL absorbs is $\min_{a', b'} \mathbb{E}_\theta [\text{KL}(\text{Bern}(m_j(\theta)) \parallel \text{Bern}(\sigma(a'(\theta - b'))))]$, minimized by Nelder–Mead. For a pair it is $\mathbb{E}_\theta [I(X_j; X_k \mid \theta)]$, computed exactly over the four response patterns. The Kullback–Leibler divergence between the joint mixture and the product of any two 2PLs splits into this mutual information plus two single-item terms, so the pair column is exactly the part that no conditionally independent model can absorb. The expected likelihood ratio of a correctly specified alternative grows as $2N$ times the per-respondent information. For mixtures this is a heuristic, because the null lies on the boundary of the parameter space [43].

B Parameters and their status

Table 10: Operational parameters of the reference design. “Provisional” means not yet derived from a calibration study. Sources: the repository’s design documents and code at the described commit.

Parameter	Value	Status and remarks
Penalties λ_b / λ_f	0.15 / 0.03	provisional; the leak depends on $(\lambda_b/\lambda_f)\sqrt{S/n}$ (Proposition 4)
Bridging threshold τ	≈ 0.08	provisional; relative to the batch (Proposition 2)
Band half-width ε	≈ 0.008	provisional
Latent dimensions d	1	design decision D31
Reviewers per item k	7–11, odd	stratified on f_u
Bootstrap refits / keep rate	10 / 0.85	pessimistic minimum, warm-started
Starts of the bridging fit	8	lowest objective kept
Minimum ratings to define the axis	30	specified, not implemented
Pilot 1 respondents	≥ 300	hypothesis
Pilot 2 respondents (latent DIF)	≥ 3000	hypothesis
Minimum DIF batch	2 items	structural (Propositions 7 and 8)
Discrimination $a_{\min}$	0.6	literature value; metric mismatch (D25)
$ b $ maximum	2.5	specified, not applied
Point-biserial minimum	0.20	literature value
Logistic DIF $ \beta_2 $ maximum	0.40	calibration pilots only
Mantel–Haenszel $ \Delta_{\text{MH}} $	1.0 / 1.5	calibration pilots only (ETS B / C)
Latent DIF gap threshold	1.0	provisional; literature 0.5; tuned to 30 anchors (Empirical finding 11)
Latent classes G	1–4	chosen by BIC
Minimum class share for the gap	0.05	—
Discount exponent α	0.5	—
Weight cap $w_{\max}$	$3 \times$ median	vacuous for $E_u \in (0, 1)$ (Proposition 15)
Evaluator slope γ	unspecified	tests use 2
Moving-average rates (up, down)	unspecified	up < down
Author prior (α_0, β_0) , decay T	(2, 3), 18 months	—
Golden-item rate η	5%	—
Probation	200 scored judgments	weight 0 before

C Reproducing the numbers

Every table and figure is generated by a script in `paper/scripts/`, which writes a CSV file to `paper/data/`. The figures and some tables read those files directly. The scripts need Python 3 with NumPy and SciPy (`scripts/requirements.txt`). `levelB_detector.py` also needs a Rust toolchain: it builds `scripts/dif-harness`, a small program that runs the production detector `scoring::dif::mixture_dif` and the protocol’s verdict rule on generated batches. All randomness is seeded.

Script	Produces	Runtime
<code>levelA_relativity.py</code>	identities of Lemma 1; Table 2	< 1 min
<code>levelA_leak.py</code>	Table 3, Figure 2	≈ 8 min
<code>levelB_information.py</code>	Table 4	seconds
<code>levelB_proxy.py</code>	Table 5	≈ 2 min
<code>levelB_detector.py</code>	Table 6; production-detector counts in the text	≈ 8 min
<code>levelB_differential.py</code>	differential gap (Section 4.5)	< 1 min
<code>levelC_bss.py</code>	Figure 3, Table 7	≈ 1 min
<code>assignment.py</code>	Table 8	seconds

`common.py` re-implements the bridging objective and the simulation's two-class DIF model in NumPy with analytic gradients and tight tolerances, so that stationarity conditions can be checked to optimizer precision. Its function `sim_levelA_dataset()` rebuilds, from the same random stream, the dataset of the repository simulation `sim/bridging_irt_dif.py`. The paper is built with `latexmk -pdf main.tex` in `paper/`.